

Stereoisomerism

Cis-trans, E/Z and optical isomers

Name _____ Class _____ Date _____

KEY IDEAS

Cis-trans: C=C with two different groups on each carbon · E/Z: rank with CIP rules (Z = together, E = opposite)

CIP: higher atomic number first · tie: move outwards, first difference · C=O counts as (O, O) · R/S: point 4 away; 1 → 2 → 3 clockwise R, anticlockwise S

Chiral centre: carbon with four different groups · enantiomers: non-superimposable mirror images

Atomic numbers: H 1 · C 6 · O 8 · Cl 17 · Br 35

Draw the molecules when it helps. Mark chiral centres with an asterisk.

A. Types of isomer

1. For each pair, say whether they are structural isomers, cis-trans isomers or the same compound: (a) butane and methylpropane (b) cis- and trans-but-2-ene (c) propan-1-ol and propan-2-ol (d) two drawings of $\text{CH}_3\text{CH}_2\text{Cl}$ rotated about the C-C bond

B. Cis-trans and E/Z

2. Which of these show cis-trans isomerism? Explain each. (a) pent-2-ene, $\text{CH}_3\text{CH}=\text{CHCH}_2\text{CH}_3$ (b) 2-methylpropene, $(\text{CH}_3)_2\text{C}=\text{CH}_2$ (c) 1,2-dibromoethene, $\text{BrCH}=\text{CHBr}$

3. Draw the cis and trans isomers of 1,2-dichloroethene. Which has the higher boiling point (60 °C or 48 °C), and why?

4. Assign E or Z: (a) $\text{CHCl}=\text{CHBr}$ with Cl and Br on the same side (b) $\text{CH}_3\text{CH}=\text{CHCl}$ with CH_3 and Cl on opposite sides

C. Optical isomerism

5. Identify any chiral centres in: (a) 2-chlorobutane, $\text{CH}_3\text{CHClCH}_2\text{CH}_3$ (b) propan-2-ol, $(\text{CH}_3)_2\text{CHOH}$ (c) lactic acid, $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$ (d) 1-chlorobutane

6. Describe how the two enantiomers of lactic acid are (a) the same (b) different. What is a racemic mixture?

D. Applications

7. Explain why a drug made as a single enantiomer can work better than the racemic mixture.

E. Challenge

8. Butan-2-ol and butan-1-ol are both $C_4H_{10}O$. Explain why butan-2-ol has two stereoisomers but butan-1-ol has none, and name the type of isomerism that relates butan-1-ol to butan-2-ol.

F. CIP rules and R/S

9. Use the CIP rules to rank each set of groups from highest to lowest priority, and explain the tie-breaks: (a) $-OH$, $-CH_3$, $-H$, $-Cl$ (b) $-COOH$, $-CH_2OH$, $-CHO$, $-CH_3$ (c) $-CH_2CH_3$, $-CH(CH_3)_2$, $-CH_2OH$

10. Lactic acid is $CH_3CH(OH)COOH$. (a) Rank the four groups on its chiral carbon. (b) In one drawing, H points away from you and, going clockwise, you meet OH, then COOH, then CH_3 . Is it R or S? (c) In another drawing, H is on a solid wedge and the path $1 \rightarrow 2 \rightarrow 3$ looks clockwise. Is it R or S?